

Sample Template for a 10-minute presentation

Problem Statement & Motivation

- *Why is this problem important?*

Problem Setup

- State variables, measurements, and noise.
- What is being estimated?
- What challenge or gap does the paper address

Method Overview

- Which estimation algorithm is used?
- One simple diagram or block figure.

Do NOT derive equations on slides — only keep the key idea.

Key Equations / Model

(Only if needed)

- State transition and measurement equations.
- Noise assumptions.
- Special features (nonlinearity, constraints, probabilistic model).

Simulation Setup

- Platform used (MATLAB, Python, ROS, C++).
- System simulated.
- Noise levels, initial conditions.
- Experiments planned (e.g., noise variations, tuning parameters).

Results: Baseline Behaviour

- Trajectory plot / state estimates.
- Error evolution (RMSE etc.).
- Covariance behaviour.

Keep graphs large and readable.

Results: Comparisons

- Effect of higher noise.
- Effect of wrong initial conditions.
- Comparison with another filter / untuned filter / naive estimator.
- Interpretation: *What does this tell us?*

Insights & Observations

- What did you learn?
- Limitations of the algorithm.
- When does it fail?
- Was the paper realistic or idealistic?

Conclusion

- One slide only.
- Summary of:
problem → method → key results → takeaways.

Backup slides may include additional plots, parameter tables, or derivations.

Evaluation rubric: Presentation Quality, Technical Depth & Understanding, Simulations, Insights, Observations & Critical Thinking, Q&A